

Tutorial 4 Fields and Energy Suggested Solutions

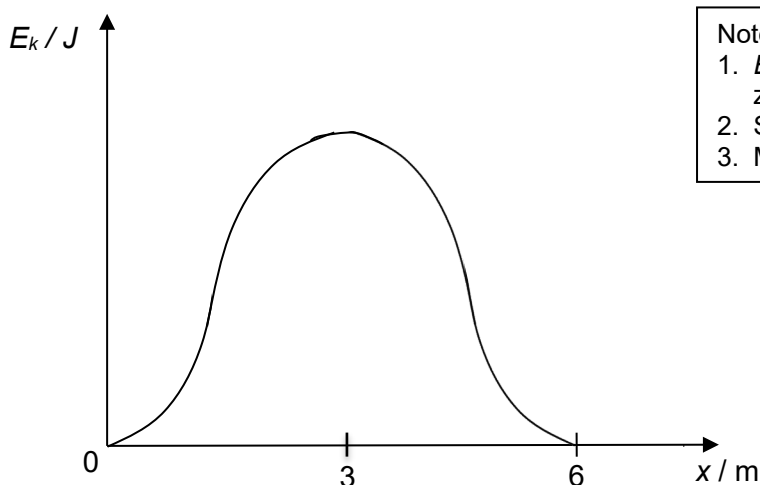
- D1** Energy is transferred through the mechanical pathway (the student's hand exerts a force on the stone) from the chemical potential store in the cells of the student's body to the kinetic store of the stone.

As the stone moves upwards until it reaches its highest point, energy is transferred mechanically from the kinetic store to the gravitational potential store of the stone. At the highest point, energy in the gravitational potential store of the stone is a maximum and there is zero energy in its kinetic store.

As the stone falls back towards the student's hand, energy is transferred mechanically from the gravitational store to the kinetic store of the stone. As there is no air resistance, there is no dissipation of energy to the internal store of the surroundings. The amount of energy in the stone's kinetic store is the same as that when it first left the student's hand.

- D2 (a) (i)** Area under the F_x vs x graph gives work done. Assuming no friction, work done by force F on the particle is converted to K.E. of the particle (motion is along x -axis, no change in G.P.E.)
- Positive work done by F_x causes an increase in K.E.
- K.E. reaches maximum at $x = 3.0$ m.
- (ii)** Area (work done) is negative from $x = 3.0$ to 6.0 m, causing the particle to lose kinetic energy. It slows down to 0 K.E. at $x = 6.0$ m (net work done on it is 0).

(b)



Note:

1. $E_k = 0$, but gradient is *not* zero at $x = 0, 6$ m
2. Steepest at $x = 1, 5$ m
3. Maximum at $x = 3$ m

From work-energy theorem, total work done by net force equals to the change in kinetic energy of the object, for this case, $\int_0^x F_x dx = E_k$. Thus F_x at each x value is equal to the *gradient of the E_k - x graph* at that x value.

D3 Note: In an *elastic* collision, the kinetic energy of the system is conserved.

(a) Elastic P.E. stored in the spring

$$= \frac{1}{2} kx^2 = \frac{1}{2} (2.0 \times 10^3) (2.0 \times 10^{-2})^2 = 0.40 \text{ J}$$

(b) Consider motion of block after collision to maximum compression of spring.

By principle of conservation of energy,

decrease in K.E. of block = increase in elastic P.E. stored in spring

$$\frac{1}{2} mv^2 - 0 = 0.40 - 0$$

$$v = \sqrt{\frac{2 \times 0.40}{1.0}} = 0.894 \text{ m s}^{-1}$$

(c) Consider motion of ball from point of release to just before collision with block.

By the principle of conservation of energy,

increase in K.E. of ball = decrease in G.P.E. of ball

$$\therefore \frac{1}{2} mv^2 - 0 = mg(L - L \cos \theta) \text{ where } L \text{ is the length of the string}$$

$$\Rightarrow (1 - \cos \theta) = \frac{v^2}{2gL} = \frac{2.68^2}{2(9.81)(1.0)}$$

$$\Rightarrow \theta = 50.7^\circ$$

D4 By principle of conservation of energy, (taking the initial position of the block as the zero G.P.E. level)

initial energy of system = final energy of system

$$(\text{Elastic P.E.} + \text{G.P.E.} + \text{K.E.})_{\text{initial}} = (\text{Elastic P.E.} + \text{G.P.E.} + \text{K.E.})_{\text{final}}$$

$$\frac{1}{2} kx^2 + 0 + 0 = 0 + mgh + 0$$

$$\frac{1}{2} (500)(0.100)^2 = 0.250(9.81)h$$

$$h = 1.019 \text{ m}$$

Hence, maximum height above the extended spring is $(1.019 - 0.100) \text{ m} = 0.919 \text{ m}$

D5 Using principle of conservation of energy,

$$E_{p \text{ initial}} + E_{k \text{ initial}} = E_{p \text{ final}} + E_{k \text{ final}} \quad (\text{assume no friction})$$

Let h be the distance of the trolley from the ground and v be the highest speed of the trolley.

$$1.5gh + 0.5g(1.0) + 0 = 1.5gh + \frac{1}{2} (1.5)v^2 + \frac{1}{2} (0.5)v^2$$

$$v^2 = 4.905$$

$$\text{Hence max. KE of trolley} = \frac{1}{2} (1.5)(4.905) = 3.7 \text{ J}$$

D6 (a) Tension in the tow bar = 2000N.

As the caravan travels at a constant velocity, the resultant force on the caravan is zero. This is only possible if the tension in the tow bar is equal in magnitude and opposite in direction to the opposing force on the caravan.

(b) Applying Newton's 2nd law to free-body diagram of the car travelling ($a = 0$),

$$F_{\text{driving}} - 2000 - 1700 = 0$$

$$F_{\text{driving}} = 3700 \text{ N}$$

$$\text{Power of car, } P = F_{\text{driving}}v = 3700 \times 20 = 74000 \text{ W} = 74 \text{ kW}$$

D7 Resistive force on car = driving force = 200 N

$$\text{Rate of work done against resistive forces} = Fv = 200 \times 20 = 4000 \text{ W}$$

$$\text{Rate of increase of GPE of the car up a gradient} = mgh/t$$

$$= 800(9.81)\left(\frac{1}{8} \times 20\right)$$

$$= 19620 \text{ W}$$

$$\begin{aligned} \text{Power needed by car} &= \text{rate of work done against resistive forces} + \text{rate of increase of GPE} \\ &= 4000 + 19620 = 23620 \text{ W} = 23.6 \text{ kW} \end{aligned}$$

OR

$$F = f + W \sin \theta$$

$$= 200 + (800)(9.81)\left(\frac{1}{8}\right)$$

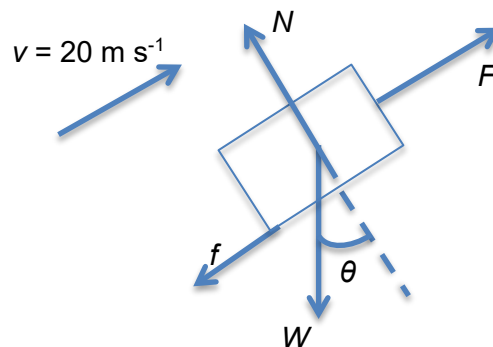
$$= 1181 \text{ N}$$

$$P = Fv$$

$$= (1181)(20)$$

$$= 23620$$

$$= 23.6 \text{ kW}$$



- D8** At constant speed, the power input by the cyclist is equal to the sum of the rate at which work is done against the resistive force and the rate of increase in gravitational potential energy:

Power input by cyclist = Power dissipated by resistive force + rate of increase of GPE

$$\text{Power input} = fv + mg(h/t) = fv + mg(d \sin \theta)/t = fv + mgv \sin \theta$$

$$\sin \theta = \frac{\text{Power input} - fv}{mgv} = \frac{0.20 \times 10^3 - 10 \times 5.0}{80 \times 9.81 \times 5.0}$$

$$\theta = 2.2^\circ$$

Note that the speed has to be converted into m s^{-1} .

OR:

Using $P = Fv$, force exerted by cyclist is $F = (0.20 \times 10^3)/(18 \times 10^3 / 3600) = 40 \text{ N}$

Since speed is constant, resultant force on the cyclist and bicycle is zero.

By Newton's second law,

$$F - mg \sin \theta - F_{\text{resistive}} = 0$$

$$40 - 80 \times 9.81 \sin \theta - 10 = 0$$

$$\theta = 2.2^\circ$$

- D9 (a)** Total change in G.P.E. of the water in 8 buckets in one revolution

$$= 8(mg\Delta h)$$

$$= 8(40 \times 9.81)(-1.6 \times 2)$$

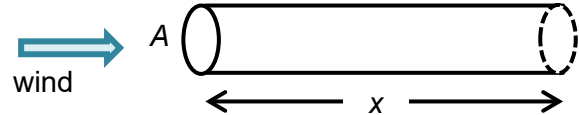
$$= -1.0 \times 10^4 \text{ J (i.e. loss in G.P.E.)}$$

- (b)** Average input power = $\frac{\text{conversion of GPE to KE in one revolution}}{\text{period}}$

$$= \frac{1.0 \times 10^4}{\left(\frac{60}{6}\right)} = 1.0 \times 10^3 = 1.0 \text{ kW}$$

- (c)** A larger number of small buckets is preferred as this would result in a more uniform torque throughout the revolution; hence the power will fluctuate less, and the rotation of the wheel would be smoother than when a few large buckets is used.

D10 Consider a tubular section of the wind flow.



$$\text{Mass of air flowing per second} = \frac{dm}{dt} = \frac{d(\rho V)}{dt} = \rho \frac{d(Ax)}{dt} = \rho A \frac{dx}{dt} = \rho Av$$

where x is the length of air column passing through the blades, hence Av is the volume flow rate, and ρ = density of air

$$\text{Kinetic energy of air flowing per unit time} = \frac{1}{2} \frac{dm}{dt} v^2 = \frac{1}{2} (\rho Av) v^2 = \frac{1}{2} \rho Av^3$$

Since overall efficiency is 40%,

$$0.40 \times \frac{1}{2} \rho Av^3 = 5000, \text{ where } A \text{ is the circular cross-sectional area of the windmill}$$

$$0.40 \times \frac{1}{2} \rho \pi L^2 v^3 = 5000, \text{ where } L \text{ is the length of the blade}$$

$$\begin{aligned} L &= \sqrt{\frac{5000}{0.40} \times \frac{2}{\pi \rho v^3}} \\ &= \sqrt{\frac{5000}{0.40} \times \frac{2}{\pi \times 1.2 \times (50000 / 3600)^3}} \\ &= 1.57 \text{ m} \end{aligned}$$

- C1** The work done dW over a small displacement dx is $T \cos \theta \cdot dx$

$$\text{Since } \cos \theta = \frac{x}{\sqrt{1.20^2 + x^2}}$$

$$W = \int_3^1 T \cos \theta \, dx = \int_3^1 -25 \frac{x}{\sqrt{1.20^2 + x^2}} \, dx$$

$$W = -25 \left[\sqrt{1.20^2 + x^2} \right]_3^1 = 41.7 \, \text{J}$$

The change in KE of the cart is 41.7 J

- C2** On a level road, power supplied by car's engine = rate of work done against friction

$$P = f v = (k v^2) v = k v^3$$

When car is moving up a slope at maximum speed $\frac{1}{2} v$, power supplied by engine is given by

$$P = (f' + mg \sin \theta) v'$$

$$k v^3 = \left(k \left(\frac{v}{2} \right)^2 + mg \sin \theta \right) \left(\frac{v}{2} \right)$$

$$k v^3 = \frac{k v^3}{8} + \frac{mg v \sin \theta}{2}$$

$$mg \sin \theta = \frac{7}{4} k v^2 \quad \dots (1)$$

When car is moving down the same slope with speed λv ,

$$P + (mg \sin \theta) v'' = (f'') v''$$

$$k v^3 + \frac{7}{4} k v^2 (\lambda v) = k (\lambda v)^2 (\lambda v)$$

$$1 + \frac{7}{4} \lambda = \lambda^3$$

$$\lambda^3 - \frac{7}{4} \lambda - 1 = 0 \text{ (shown)}$$